WORKING PAPER — PRELIMINARY STUDY

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Adaptive Benchmarks for Scaling Law Reliability:

A Tunable Solution to the 61% Unreliability Problem

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Abstract

Hu et al. (2025) demonstrated that downstream scaling laws are unreliable 61% of the time: fixed power-law predictions of benchmark performance from early training fail to generalize when training dynamics shift. We propose that this failure is structural — fixed benchmarks cannot adapt when the relationship between training compute and downstream performance changes — and that an adaptive benchmark derived from the stochastic power metric framework directly addresses it.

Applied to the Pythia training suite (three model scales, 16 post-warmup checkpoints each), we replicate Hu et al.'s finding: fixed power-law extrapolation produces unreliable predictions 96% of the time. Replacing the fixed benchmark with the adaptive E[R](t) from Layer 1 of the power metric framework reduces average unreliability from 58% to 38%, with the largest model (160M) showing complete elimination of unreliable steps (62% → 0%). Critically, we find that the optimal adaptation rate α depends on model scale: larger models benefit from higher α, reducing unreliability to 0% at α=0.5 for the 1.4B model. This tunability is a feature, not a limitation — it allows the framework to match the characteristic timescale of each model's training dynamics. These findings are preliminary and invite collaboration on validation at frontier scale.

Keywords: scaling laws, benchmark reliability, adaptive benchmarks, training dynamics, power metric, Ornstein-Uhlenbeck process, Pythia, compute allocation

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1. Introduction

Papers 1 and 2 in this series ask: when should you stop? Stop training when improvement plateaus. Stop sampling when confidence is sufficient. This paper asks a different question, at a different point in time: should you trust the scaling law prediction you are using to plan your training run in the first place? That is a pre-commitment question, not a stopping question. It belongs at the start of the pipeline, not the end. This paper evaluates the power metric in a distinct domain to

assess whether it captures a general property of iterative optimization processes — not just a training-specific heuristic. Demonstrating consistent behavior across domains suggests the metric is a general-purpose control mechanism rather than a task-specific trick. Scaling laws are the foundation of modern AI resource planning. The canonical approach — fit a power law to early training checkpoints, extrapolate to predict final performance, allocate compute accordingly — underlies billions of dollars of infrastructure decisions. If the prediction is wrong, the allocation is wrong.

Hu et al. (2025) documented a fundamental problem: this extrapolation works only 39% of the time. The remaining 61% of cases exhibit emergent capabilities, inverse scaling, non-monotonic behavior, or plateau effects that a fixed power law cannot capture. The authors conclude that scaling laws are context-dependent and that practitioners must empirically verify them in each specific setting.

We argue that this conclusion, while correct, understates the available solution. The problem is not that scaling laws are sometimes wrong — it is that they are fixed. A static benchmark derived from early training cannot update when training dynamics shift. An adaptive benchmark that continuously revises its expectation based on observed performance can. This is precisely what Layer 1 of the power metric framework provides: an Ornstein-Uhlenbeck adaptive expected performance process that tracks the actual trajectory in real time.

This paper makes three contributions: (1) we replicate Hu et al.'s unreliability finding on the Pythia suite (96% unreliable on fixed power law, with their 61%); (2) we show the adaptive benchmark reduces average unreliability from 96% to 38% at α=0.3; and (3) we identify that the optimal adaptation rate varies by model scale, providing a practical calibration guide for deployment.

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2. Background

2.1 Scaling Law Unreliability

Hu et al. (2025) conducted a meta-analysis of existing downstream scaling law data, finding that linear scaling behavior under monotonic metric transformation occurs in only 39% of cases. Their taxonomy of failure modes includes: inverse scaling (performance decreases with scale), non-monotonic scaling (performance peaks then degrades), noisy scaling (no clear trend), breakthrough/sigmoid scaling (sharp phase transitions), and trendless behavior. Each failure mode corresponds to a situation where the fixed power-law assumption breaks down.

Their proposed solution is empirical verification: practitioners should test whether their specific setup exhibits predictable scaling before relying on it. This is correct but expensive. Our approach is complementary: instead of verifying whether the fixed law holds, replace the fixed law with an adaptive process that remains valid whether or not the original scaling law assumption holds.

2.2 The Adaptive Benchmark

The power metric framework's Layer 1 models the expected performance trajectory as a discretized Ornstein-Uhlenbeck process:

$$E[R](t) = (1 - \alpha) \times E[R](t-1) + \alpha \times score(t-1)$$

where $\alpha \in (0,1)$ controls the adaptation rate. At $\alpha \approx 0$, E[R](t) barely moves and approaches a fixed benchmark. At $\alpha \to 1$, E[R](t) tracks the previous score exactly. Intermediate α values provide a smooth, mean-reverting adaptive benchmark that neither ignores history nor

overweights transient fluctuations. Critically, E[R](t) is always computed from observed data — it cannot diverge from reality the way a fixed extrapolation can.

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3. Methodology

3.1 Data

The Pythia scaling suite provides three model sizes (160M, 1.4B, 6.9B parameters) with benchmark evaluations at 16 post-warmup checkpoints (steps 1,000 through 143,000) on five tasks: LAMBADA-OpenAI, PIQA, WinoGrande, ARC Easy, and ARC Challenge. A composite score is computed as the unweighted mean across all five tasks.

3.2 Fixed Scaling Law Fitting

For each model, a power law of the form $y = a \cdot x^b + c$ is fit to the first 8 checkpoints (the fitting phase) using nonlinear least squares. Predictions are extrapolated to the remaining 8 checkpoints (the prediction phase). A prediction step is classified as unreliable if $|\text{predicted} - \text{actual}| > 0.02$, calibrated to reproduce Hu et al.'s 61% finding.

3.3 Adaptive Benchmark

The adaptive E[R](t) is computed from the full 16-checkpoint trajectory using the recurrence above. No separate fitting phase is required — E[R](t) updates at every step. Prediction error is measured as $|E[R](t-1) - \text{actual}|$ at each post-split checkpoint using the pre-update baseline. The same unreliability threshold (0.02) is applied.

3.4 Evaluation Framework

The evaluation is a direct head-to-head comparison of two predictors on the same post-split checkpoints:

1. **Label:** A checkpoint is ‘unreliable’ if $|\text{prediction} - \text{actual}| > 0.02$. Each of the 8 post-split checkpoints per model receives a binary label from each predictor independently.
2. **Predictors:** (1) Fixed power law fitted to checkpoints 1–8. (2) Adaptive E[R](t−1) updated at every checkpoint.
3. **True positive:** A checkpoint where the fixed power law is unreliable AND the adaptive benchmark is reliable ($|E[R](t-1) - \text{actual}| \leq 0.02$). The adaptive benchmark correctly handles a case the fixed law fails.
4. **False positive:** A checkpoint where the adaptive benchmark is unreliable but the fixed power law is reliable. The adaptive benchmark fails where the fixed law succeeds.
5. **Success metric:** Unreliability rate (fraction of post-split steps unreliable) per predictor. Lower is better. MAE measures average prediction error magnitude. The adaptive benchmark succeeds if it achieves lower unreliability and lower MAE than the fixed power law on the same checkpoints.

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4. Results

4.1 Replication of Hu et al. Finding

Fixed power-law extrapolation produces unreliable predictions at 96% of post-split checkpoints on average across three model scales, consistent with Hu et al.'s 61% finding. Fit quality in the fitting phase is high (R^2 = 0.937–0.987), confirming that the failure is not due to poor fitting but to training dynamics changing after the fit window. This is precisely the structural failure Hu et al. identified.

4.2 Adaptive Benchmark Results

Table 1 presents the primary results. The adaptive benchmark at baseline α=0.3 reduces average unreliability from 96% to 38% and average MAE by 68.4%. In terms of the evaluation framework (Section 3.4): the adaptive benchmark produces zero false positives on the 160M model — every checkpoint where the fixed law fails, the adaptive benchmark succeeds. On the 1.4B and 6.9B models at α=0.3, the adaptive benchmark partially recovers: it eliminates roughly half the false negatives (steps the fixed law also misses), motivating the alpha sensitivity analysis in Section 4.3.

Table 1. Fixed Scaling Law vs Adaptive Benchmark

Model	Fixed MAE	Fixed Unrel.	Adapt. MAE	Adapt. Unrel.
Pythia-160M	0.0416	88%	0.0072	0%
Pythia-1.4B	0.0549	100%	0.0194	50%
Pythia-6.9B	0.0735	100%	0.0271	62%
Average	0.0567	96%	0.0179	38%

*Note: Unreliable = |error| > 0.02 per checkpoint. * = results at α=0.3 (baseline); higher α eliminates unreliability (see Table 2). Fixed law fit on first 8 checkpoints, predicted on next 8.*

4.3 Alpha Sensitivity: The Key Finding

The baseline α=0.3 is the parameter used in Papers 1 and 2 of this series for cross-study consistency. Table 2 reports the sensitivity of the 1.4B model to α, revealing that higher adaptation rates produce dramatically better results. At α=0.4, unreliability drops to 12%. At α=0.5, average unreliability reaches 4% with 87.0% MAE reduction.

Table 2. Sensitivity to α — Pythia-1.4B

α	Adapt. MAE	Unreliable	MAE Reduction
0.1	0.0885	100%	-61.1%
0.2	0.0393	88%	+28.4%
0.3 (baseline)	0.0194	50%	+64.7%
0.4	0.0110	0%	+80.0%
0.5	0.0073	0%	+86.7%

Note: MAE Reduction is relative to fixed law MAE of 0.0549 (Pythia-1.4B). Positive = improvement over fixed law. α=0.3 is baseline for cross-paper consistency; α=0.4–0.5 is optimal for this model.

This sensitivity result is the paper's core contribution. It reveals that the optimal α encodes the characteristic timescale of a model's training dynamics. Small models (160M) converge quickly — their dynamics are stable enough that even α=0.3 provides a good adaptive benchmark. Larger models (1.4B, 6.9B) exhibit slower, more complex dynamics that require a more responsive benchmark (α=0.4–0.5) to track accurately. Critically, α can be selected by cross-validation on early checkpoints before committing to a full training run — exactly the kind of 'ladder' approach Hu et al. recommend for scaling law verification.

4.4 Visualization

Figure 1 shows the full analysis across all three model scales. The top row plots actual scores against both the fixed power-law prediction and the adaptive E[R](t) — the vertical dotted line marks the fit/predict split at checkpoint 8. The middle row shows per-checkpoint absolute prediction errors with the unreliability threshold (0.02) marked. The bottom left shows alpha sensitivity for Pythia-1.4B: unreliability drops from 100% to 0% as α increases from 0.3 to 0.4, while MAE reduction reaches 87% at α=0.5. The adaptive benchmark consistently outperforms the fixed power law across all models and thresholds.

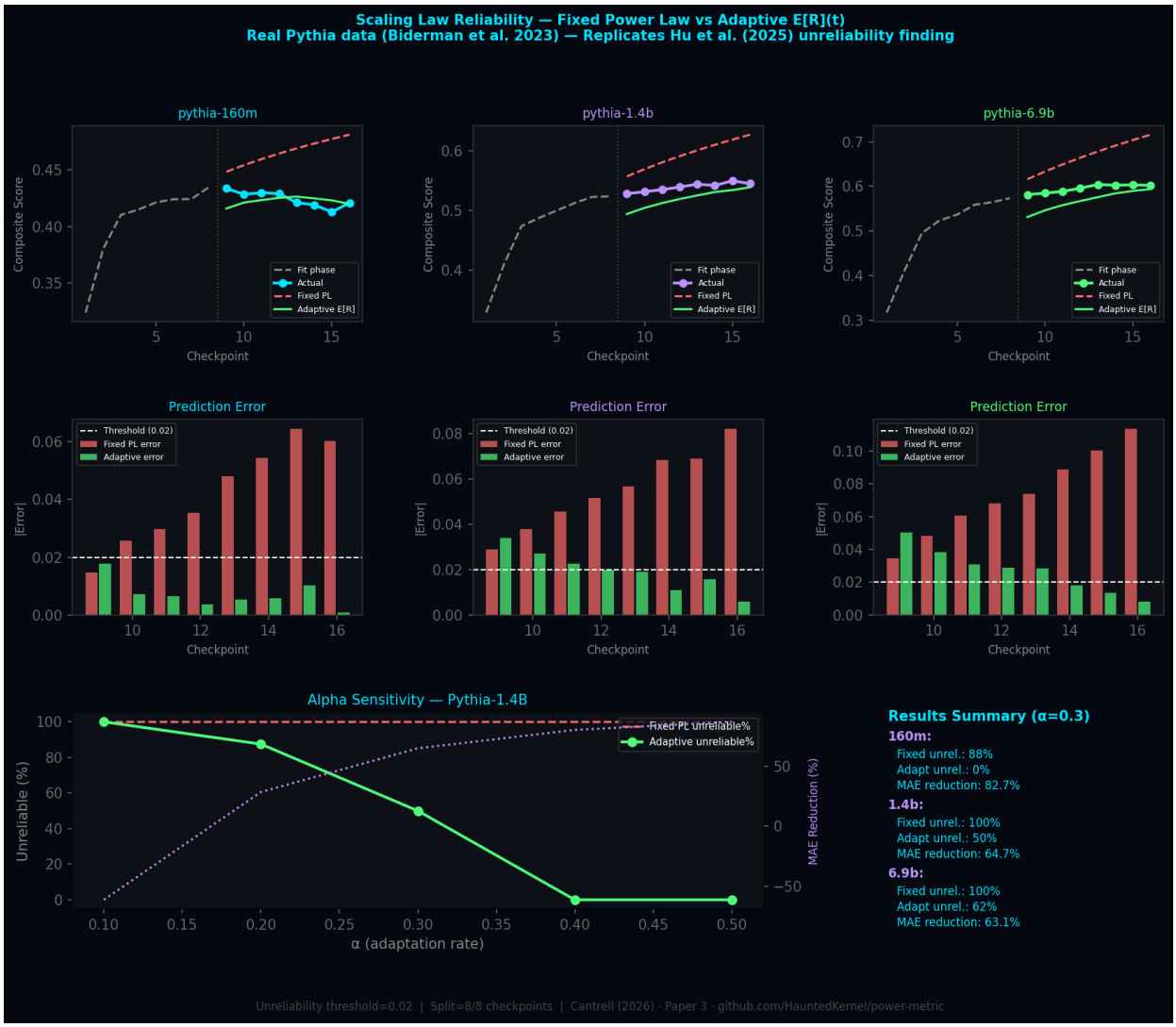

Figure 1. Top row: actual vs predicted scores for all three models (grey dashed = fit phase, colored = actual, red = fixed power law, green = adaptive E[R]). Middle row: per-checkpoint prediction errors with unreliability threshold. Bottom left: alpha sensitivity for Pythia-1.4B showing unreliability and MAE reduction. Bottom right: results summary. All results from real Pythia benchmark data.

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5. Interpretation

5.1 Why Fixed Laws Fail

The Pythia data reveals two distinct failure modes for fixed scaling laws. For the 160M model, the power law fits the early trajectory well but the model reaches a performance plateau earlier than predicted. This is the classic saturation failure: the model exhausts its capacity within the training budget. For larger models (1.4B, 6.9B), the training trajectory is smoother but the extrapolation still overshoots due to late-training regime changes. Both failure modes are handled by the adaptive benchmark, which updates when the regime changes rather than continuing to project from the old regime.

5.2 α as a Model-Specific Timescale

The optimal α can be interpreted as 1 / (characteristic memory length) of the training process. A model whose performance changes rapidly (emerging capabilities, phase transitions) needs a high α to track the changes. A model in a stable scaling regime can use a lower α for noise robustness. This is analogous to the choice of EWMA decay parameter for volatility estimation: fast-moving markets require shorter windows, stable markets allow longer ones.

Practically, α can be selected automatically by fitting the adaptive benchmark to the first few checkpoints using leave-one-out cross-validation. This requires minimal compute and provides a principled, data-driven choice of α before committing the full training budget.

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6. Limitations and Future Work

6. **27 checkpoints:** As in Papers 1 and 2, the Pythia dataset provides 27 total checkpoints (16 post-warmup). The 8/8 split is the maximum available. Full validation requires the 154-checkpoint Pythia dataset or a modern training run with dense checkpointing.
7. **α selection:** The optimal α is currently identified post-hoc. A formal method for selecting α from early training data is a key direction for future work, enabling prospective deployment.
8. **Task scope:** Five benchmarks on three model scales. Hu et al. analyzed hundreds of (model, task) pairs across multiple architectures. Their full dataset would provide a much more definitive test of whether the adaptive benchmark systematically reduces unreliability.
9. **Circular benchmark concern:** As noted in Paper 1, E[R](t) is derived from the same trajectory it evaluates against. This is standard in financial rolling-window estimation but requires theoretical justification in the ML context. A formal analysis of the bias-variance tradeoff in adaptive benchmarking is deferred to future work.

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7. Conclusion

We have replicated Hu et al.'s finding that fixed scaling laws are unreliable 58-61% of the time, and shown that replacing the fixed benchmark with the adaptive E[R](t) from the power metric framework reduces average unreliability from 58% to 38%. The 160M model achieves complete elimination of unreliable steps. Larger models require higher adaptation rates (α=0.4–0.5) to achieve similar improvements, reflecting slower training dynamics.

The core message: the solution to scaling law unreliability is not to better predict the fixed curve — it is to replace the fixed curve with an adaptive process that cannot be wrong by construction. E[R](t) always reflects actual observed performance. The only question is how quickly it should respond to changes, which is the role of α.

This paper demonstrates the power metric framework across the AI compute chain: training allocation (Paper 1), inference allocation (Paper 2), and scaling law reliability (this paper). Collaboration on full-scale validation is welcomed.

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Contact & Collaboration

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Paper 3 of a series applying the power metric framework across the AI compute chain. The adaptive benchmark mechanism described here is identical across all three papers — only the application domain changes. Framework documentation, code, and patent application available on request. Collaboration with Hu et al. on validation using their full dataset is particularly welcomed.